Sushant Kamble

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Engineer and XR researcher with an M.S. in Information Studies from UT Austin and over six years of industry experience in software engineering and interaction design. Currently a UX Engineer at Beheld Health, pursuing independent research in wearable haptics, multimodal interaction, and spatial interfaces to make immersive experiences more intuitive and accessible.

EDUCATION

The University of Texas at Austin **Austin, TX**
Master of Science in Information Studies Aug 2022 – May 2024

- Academic focus: Human-Computer Interaction, immersive technologies, and user-centered evaluation.
- Relevant project work: Immersive VR interactions, WebXR prototyping, and spatial interface design.

Savitribai Phule Pune University **Pune, India**
Bachelor of Engineering in Information Technology Jul 2015 – Jun 2018

- Academic focus: Mobile and Web application development

RESEARCH EXPERIENCE

The University of Texas at Austin **Austin, TX**
Graduate Research Assistant (Advisor: Shefali V. Patil) 2023 – 2024

- Developed automated data-acquisition pipelines using Node.js and Selenium, collecting and processing information from Wikipedia, IMDb, and GDELT to support research on media and information analysis.
- Deployed and ran LLaMA-2 models locally for research data analysis.
- Developed customized Qualtrics survey interfaces using JavaScript, implementing conditional branching, eligibility screening, and automated disqualification logic for large-scale participant studies.

RESEARCH AND PROTOTYPING PROJECTS

Phantom Touch: Multimodal Wearable Interface for Immersive VR
Independent Research Project (In Progress) 2026 – Present

Technologies: Unity, C#, Meta Quest 3, ESP32-S3, ERM/LRA Actuators

- Investigating wearable haptics for virtual surface perception, placing actuators on the backs of fingers to preserve direct physical interaction.
- Developing a pipeline connecting Meta Quest hand tracking and Unity touch events to ESP32-S3 wearable actuators via Bluetooth Low Energy.
- Plan to explore movement-dependent texture rendering using virtual surface maps and linear resonant actuators.
- Planning perceptual evaluations to examine how texture detail, finger movement, tracking uncertainty, and actuator response affect texture discrimination.

Elixir Expedition: Action-Dependent Perception and Environmental Awareness in Immersive VR
Academic XR Research Project | UT Austin | Four-Person Team 2023

Technologies: Unity, C#, Meta Quest 3, XR Interaction Toolkit, Blender

- Co-developed an ancient Egyptian VR experience exploring perceptual awareness, change blindness, and embodied interaction.
- Led experimental design, Unity prototyping, and performance optimization for standalone Meta Quest hardware.
- Implemented action-driven interactions and environmental changes triggered by head orientation and proximity.
- Conducted an exploratory evaluation with eight participants, gathering feedback on awareness and recall of environmental changes.

Solar-Assisted Smartphone Charging Case
Undergraduate Hardware and Entrepreneurship Project | Two-Person Team 2018

Technologies: Arduino, Solar Energy Harvesting, Supercapacitors, Android

- Co-developed a smartphone charging case integrating solar energy harvesting, supercapacitor storage, and regulated battery charging.
- Prototyped circuit and enclosure configurations to balance energy efficiency, portability, and form factor.
- Built a companion Android application for charging control and configurable charge limits.
- Registered the project as a startup venture through the university's Technology Business Incubation Center, receiving mentorship and research funding through the Make in India initiative.

INDUSTRY EXPERIENCE

Beheld Health **Austin, TX**
Frontend / User Experience Engineer Jul 2024 – Present

- Lead frontend architecture and development of a React Native healthcare app supporting participant outreach, scheduling, communication, and documentation.
- Designed scheduling and task-management interfaces with prioritization and follow-up workflows.
- Integrated Twilio communication features and built reusable components for iOS and Android.
- Improved image-loading responsiveness by approximately 40% through frontend optimization.
- Automated NPS surveys and reminders, implemented CI/CD pipelines, and integrated error monitoring.
- Collaborated with healthcare professionals and researchers to develop user-centered mobile interfaces.

Scalex Cloud **Pune, India**
Software Engineer II Jul 2018 – Jul 2022

Client: Clarista

- Led frontend development and team delivery for Clarista's visual data platform.
- Built Flows, a no-code editor using React, Redux, and D3.js, enabling drag-and-drop data transformation.
- Developed data cataloging, lineage visualization, and a real-time analytics dashboard builder.
- Established reusable components and a shared design system to improve consistency and development efficiency.
- Collaborated across design and engineering on architecture, prototyping, usability, and feature delivery.

Client: Microsoft Research

- Developed a dynamic React survey deployed on Amazon Mechanical Turk to collect text classification data for training Cortana models in collaboration with Microsoft Research.

TALKS AND WORKSHOPS

Unity Optimization for Meta Quest **Austin, TX**
Technical Talk | Virtual Environments, UT Austin 2023

- Delivered a technical session on optimizing Unity-based VR applications for standalone Meta Quest headsets, emphasizing rendering performance, frame-rate stability, and user comfort.
- Covered interpupillary distance calibration, occlusion culling, static batching, Level of Detail (LOD), baked lighting and shadows, and light and reflection probes.
- Demonstrated dynamic switching between precomputed lightmaps to support interactive lighting changes without expensive real-time lighting updates.
- Guided participants through performance monitoring and bottleneck identification using Meta Quest Developer Hub and OVR Metrics Tool.

Blending Realities: Introductory XR Workshop **Austin, TX**
Bezi Fellow | Workshop Co-Organizer and Facilitator *2024*

- Co-organized and facilitated a beginner-friendly AR/VR/MR prototyping workshop in collaboration with Bezi as part of a five-person team.
- Selected browser-based XR development tools to lower technical barriers and make immersive prototyping accessible without requiring complex software installations.
- Developed interactive demonstrations introducing 3D interaction design, WebXR, and texture optimization.
- Supported participants through hands-on prototyping and experimentation.
- Prepared and shared post-workshop learning resources through Notion to support continued independent exploration of XR technologies.

SKILLS

XR Development	Unity, XR Interaction Toolkit, WebXR, Three.js, Meta Interaction SDK
3D & Spatial Design	Blender, Figma, Spatial UI Design (Bezi), VR Performance Optimization
Embedded Systems & Physical Computing	Embedded Programming, Sensor Integration, Interactive Hardware, Digital Fabrication, Physical Prototyping
Programming & Frontend	JavaScript, TypeScript, Python, C#, React, React Native, Redux, Next.js, HTML, CSS
Backend & Infrastructure	Node.js, Express.js, REST APIs, AWS Lambda, Firebase, MongoDB, MySQL, Git
Research & Evaluation	Experimental Prototyping, User Studies, Qualtrics, Heuristic Evaluation, Interaction Design, Data Visualization, Automated Data Collection
Development, Testing & Monitoring	CI/CD, Build and Deployment Automation, Error Monitoring, Browser Automation, XR Performance Profiling